Automated Changelog Generator with Analytics Dashboard

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Automating changelogs with analytics, visualization, and containerized deployment

Problem Statement

Time-Consuming

Manual changelog maintenance requires hours of documentation time that could be spent on development

Inconsistent

Developers often forget or skip documenting changes, leading to incomplete and unreliable changelogs

Low Visibility

Lack of clear visibility into commit history and overall development activity across the team

Solution Overview

01

Extract Git History

Pull commit data from your Git repository automatically

02

Parse & Categorize

Identify conventional commit prefixes (feat, fix, docs)

03

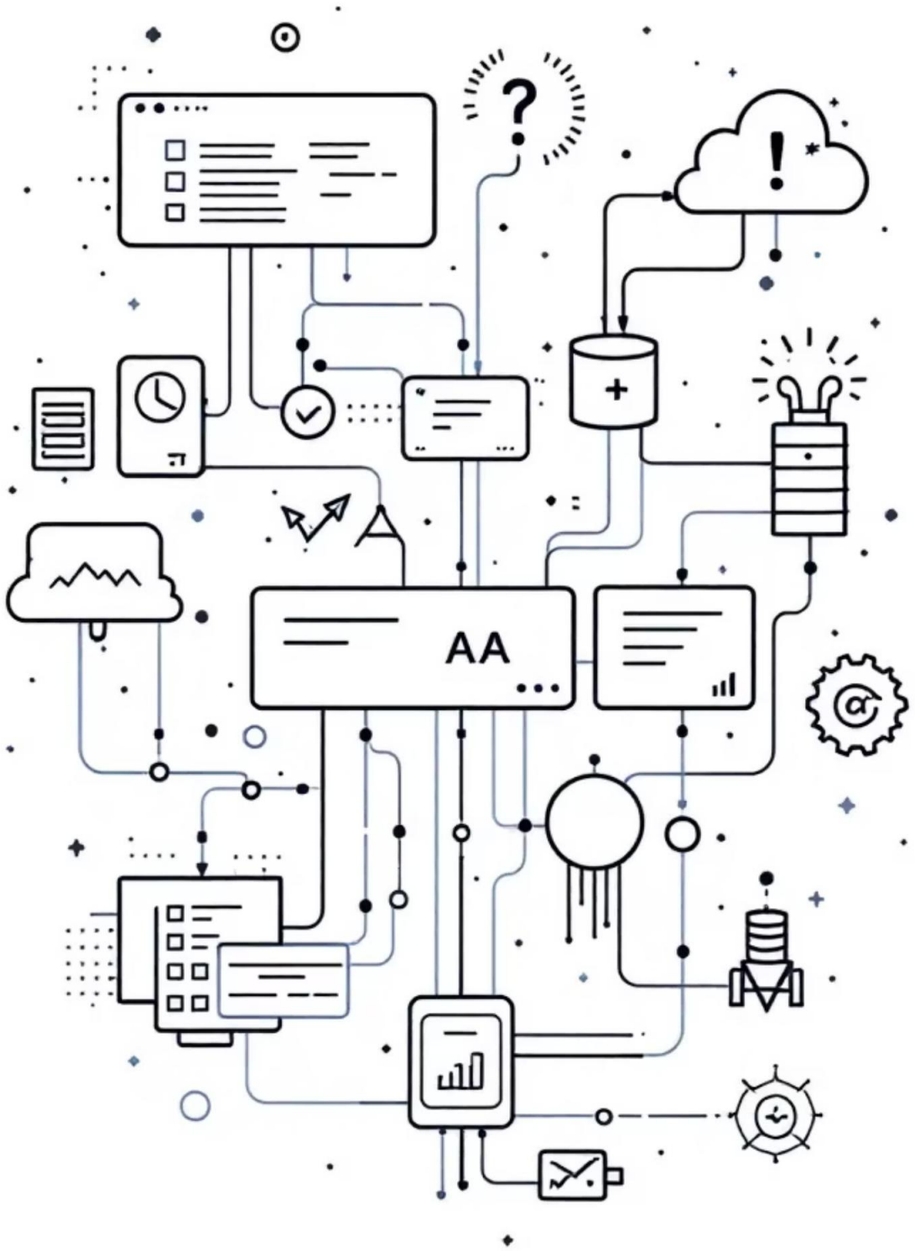
Generate Changelog

Create structured Markdown changelog with organized sections

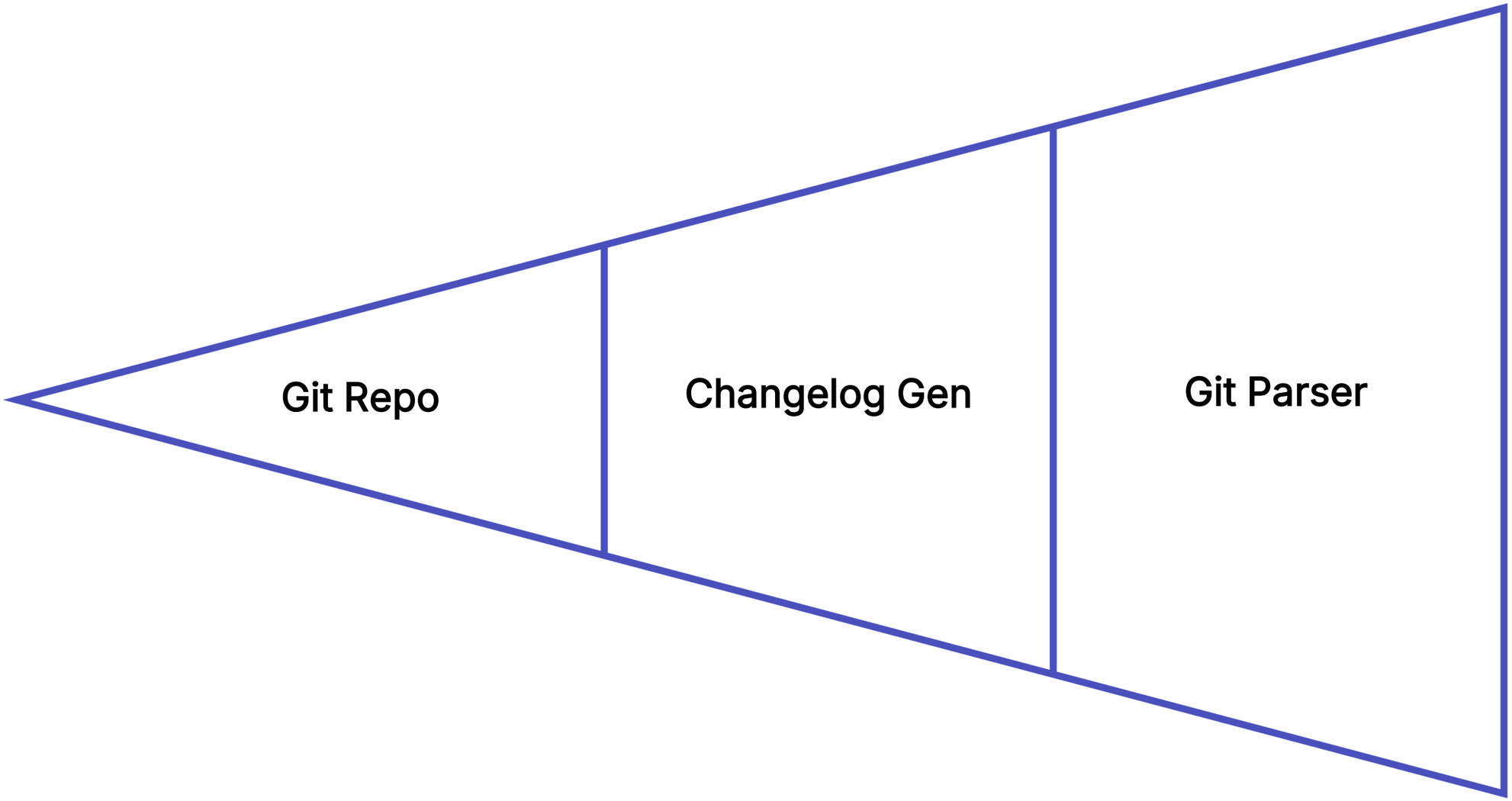
04

Visualize Data

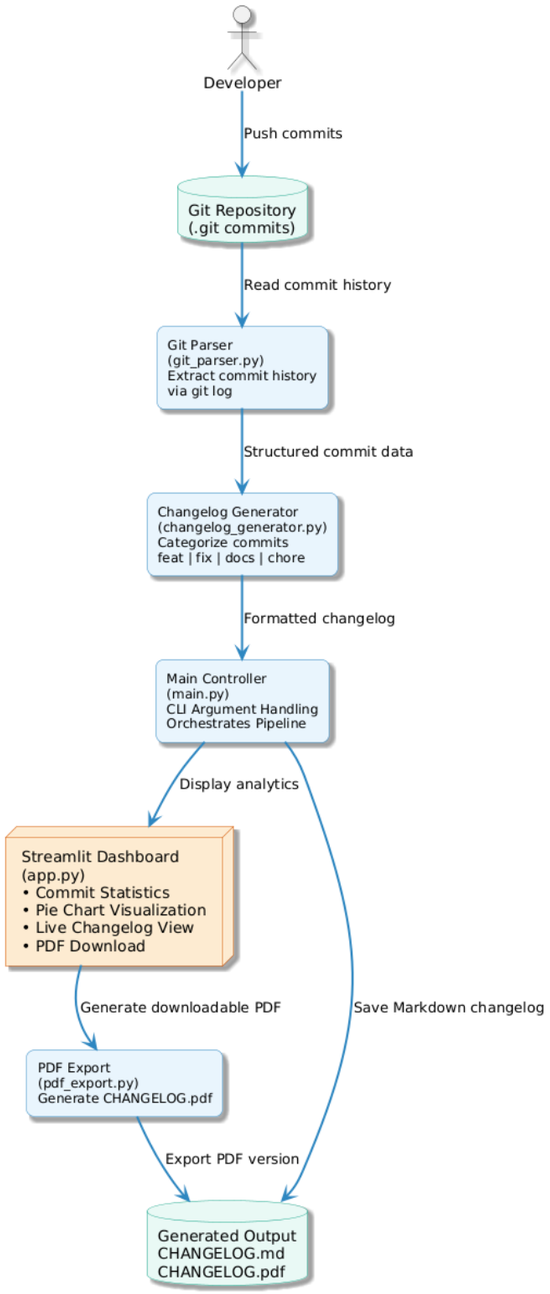
Interactive Streamlit dashboard with real-time analytics



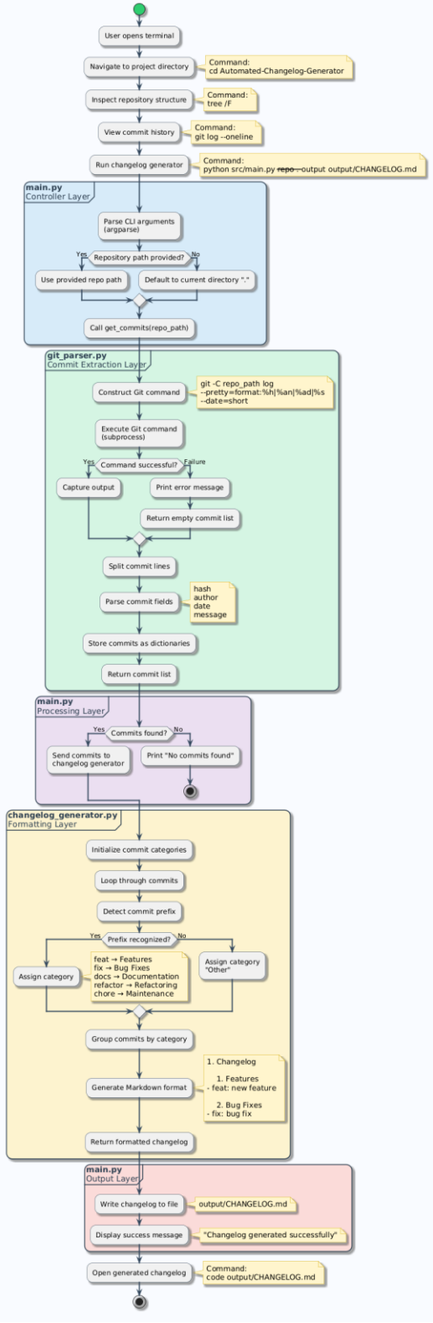
System Architecture



Automated Changelog Generator Pipeline



Automated Changelog Generator - Workflow Pipeline



Features & Enhancements

Smart Categorization

Automatic classification of commits by type: feat, fix, docs, style, refactor, test, chore

Real-Time Generation

Changelog updates instantly as new commits are pushed to the repository

Visual Analytics

Pie charts show commit distribution across categories and time periods

PDF Export

Generate professional PDF changelogs for release documentation and sharing

Merge Detection

Track branch integration and workflow activity through merge commit analysis

DevOps Implementation & Impact

Git Version Control

Full commit history tracking and repository management using Git workflows

Documentation Automation

Eliminated manual changelog maintenance, freeing developer time for core work

Containerization

Docker provides isolated, reproducible environment for consistent deployment

Scalable Architecture

Modular design supports growth from small projects to enterprise repositories

Future Scope

- CI/CD pipeline integration with GitHub Actions
- Automated release notes generation
- Integration with project management tools